

Tutorial Week 7: Bayes Theorem

Part 1: Descriptive

1. Given HMM model represented by $M = (A, B, \pi)$. Describe the elements of a HMM.
2. Describe the forward recursion algorithm.

Part 2:

The below Multiple-Choice Questions (MCQs) focuses on “Hidden Markov Model”.

1. What are the possible values of the variable?
 - a) Variables
 - b) Literals
 - c) Discrete variable
 - d) Possible states of the world
2. Where does the Hidden Markov Model is used?
 - a) Speech recognition
 - b) Understanding of real world
 - c) Both speech recognition and understanding of real world
 - d) None of the mentioned

Part 3: Markov Model of Weather

Consider the weather condition is observable once a day Once a day as one of

S_1 : rainy S_2 : cloudy S_3 : sunny. The state transition probabilities A is:

$$A = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.1 & 0.8 \end{bmatrix}$$

1. Given the weather on day 1 ($t = 1$) is sunny(S_3), what is probability that the weather for the next 7 days will be sun-sun-rain-rain-sun-cloudy-sun?
2. Given that the model is state i , what is the probability that it stays in state i for exactly d days? What is the expected duration in state i (also conditioned on starting in state i)?